

NumPy — LEVEL 2: NumPy Array Basics

Beginner-friendly notes • Line-by-line

2. NumPy Array Basics

1. What is an array?

An array is a collection of values stored together. In NumPy, arrays are mainly used to store and process numerical data efficiently.

2. 1D Array

A 1D array is like a single row of values. Example: [10, 20, 30, 40].

3. 2D Array

A 2D array contains rows and columns, similar to a table. Example: [[1, 2], [3, 4]].

4. 3D Array

A 3D array contains multiple 2D arrays. You can think of it as a collection of tables.

5. N-Dimensional Array

An N-dimensional array can have any number of dimensions supported by NumPy. The number of dimensions is represented by ndim.

6. Creating an array

The easiest way to create a NumPy array is to use `np.array()` with a Python list or nested lists.

7. `np.array()`

`np.array()` converts a Python list or other sequence into a NumPy array.

Example:

```
import numpy as np

arr = np.array([10, 20, 30, 40])
```

8. Array indexing

Indexing means accessing a specific element of an array. NumPy uses zero-based indexing, so the first element has index 0.

Example:

`arr[0]` gives 10 when `arr = [10, 20, 30, 40]`.

9. Array slicing

Slicing means selecting a range of elements from an array.

Example:

`arr[1:4]` selects elements from index 1 up to, but not including, index 4.

10. Array shape

`shape` tells you the size of the array along each dimension. For example, a 2×3 array has shape (2, 3).

11. Array dimensions

`ndim` tells you how many dimensions an array has. A 1D array has `ndim = 1`, a 2D array has `ndim = 2`, and so on.

12. Array size

`size` tells you the total number of elements in the array.

13. Array data type

`dtype` tells you the data type of the elements stored in the array, such as `int64` or `float64`.

14. Array item size

`itemsize` tells you how many bytes are used to store one element of the array.

15. Number of bytes

`nbytes` tells you the total number of bytes used by all elements of the NumPy array.

16. Array memory layout

NumPy stores array data in a structured block of memory. The array's strides describe how many bytes NumPy moves in memory to reach the next element along each dimension.

Important properties to remember:

`arr.shape` → tells the size of each dimension

`arr.ndim` → tells the number of dimensions

`arr.size` → tells the total number of elements

`arr.dtype` → tells the data type

`arr.itemsize` → tells bytes used by one element

`arr.nbytes` → tells total bytes used by array

`arr.strides` → describes the memory steps between elements

Beginner Tip: Practice 1D and 2D arrays first. Understand indexing, slicing, `shape`, `ndim`, `size`, `dtype`, `itemsize`, and `nbytes` before moving to advanced NumPy topics.